Telecom Customer Churn Analysis Report

Project Title
Telecom Customer Churn Investigation using Python – Exploratory Data Analysis (EDA)

Project Type
Data Analytics Portfolio Project

Domain
Telecommunications

Primary Objective
Identify customer churn patterns through exploratory data analysis to understand factors associated with customer attrition and support business decisions aimed at improving customer retention.

Primary Tool

- Python

Python Libraries Verified from the Uploaded Project

- pandas
- matplotlib

Dataset

- WA_Fn-UseC_-Telco-Customer-Churn.csv

Project Files Verified

- telecom_churn.py
- Telecom_Churn_Analysis_Report.pdf
- WA_Fn-UseC_-Telco-Customer-Churn.csv

Purpose of this Document

This document serves as the permanent technical and business documentation for the Telecom Customer Churn Investigation project. Its objective is to preserve every verified project detail so the project can be understood, explained, maintained, and reused without reopening the original files.

The document is intended to function as the single source of truth for this project. It captures the business objective, dataset characteristics, analytical workflow, implementation decisions, business interpretations, technical approach, and portfolio value. Future deliverables such as resumes, interview responses, GitHub documentation, portfolio summaries, and presentation material can all be generated directly from this documentation.

Only information verified from the uploaded project is documented. Any information that cannot be confirmed from the uploaded files will be explicitly identified as unavailable rather than assumed.

Revision History

Version	Description	Status
1.0	Initial documentation created from uploaded Python project and dataset	Current

No previous revision history was available in the uploaded project.

Table of Contents

Project Identity

Project Name

Telecom Customer Churn Investigation using Python

Project Category

Exploratory Data Analysis (EDA)

Business Domain

Telecommunications

Programming Language

Python

Verified Python Libraries

- pandas
- matplotlib

No additional Python libraries were present in the final uploaded script.

Dataset

WA_Fn-UseC_-Telco-Customer-Churn.csv

Dataset Size (Verified)

- Rows: 7,043
- Columns: 21

Target Variable

Churn

Possible values observed in the project:

- Yes
- No

Primary Analytical Objective

To investigate customer churn patterns by exploring relationships between customer demographics, service characteristics, contract types, billing information, tenure, and churn status.

Executive Summary

Customer churn is one of the most important performance indicators for subscription-based businesses because retaining existing customers is generally more cost-effective than acquiring new ones. This project performs exploratory data analysis on a telecom customer dataset to identify characteristics commonly associated with customer attrition.

The uploaded Python implementation follows a structured analytical workflow beginning with dataset loading, inspection, validation of missing values, conversion of the TotalCharges field into a numeric data type, removal of invalid records, and generation of business-focused visualizations using matplotlib.

Five analytical visualizations were created in the project:

- Customer Churn Distribution
- Churn by Gender
- Tenure vs Churn
- Monthly Charges vs Churn
- Contract Type vs Churn

The project also calculates the overall churn rate from the cleaned dataset.

Rather than building a predictive machine learning model, the project focuses on descriptive analytics, enabling business users to understand existing customer behavior and identify areas where retention strategies could be improved.

The analysis particularly emphasizes customer tenure, contract duration, pricing, and demographic segmentation as important dimensions for understanding churn behavior.

Business Background

Telecommunication companies operate in a highly competitive environment where customers have numerous alternatives. Customer acquisition requires substantial investment in marketing, promotions, installation, and customer onboarding. Consequently, losing existing customers directly impacts revenue growth and increases operating costs.

Understanding why customers discontinue services enables organizations to improve retention strategies, strengthen customer relationships, optimize pricing plans, and design more effective service offerings.

The uploaded project investigates churn using historical customer information containing demographic attributes, subscription characteristics, contract information, payment methods, service usage, billing amounts, and customer tenure.

By examining these variables individually against customer churn, the project aims to identify meaningful behavioral patterns that can support business decision-making.

The project is exploratory rather than predictive, making it suitable for initial business investigation before developing advanced predictive churn models.

Problem Statement

The primary business problem addressed in this project is understanding the characteristics of customers who are more likely to discontinue telecom services.

The uploaded Python analysis attempts to answer this problem by examining customer information across multiple business dimensions and comparing churned customers with retained customers.

The project specifically investigates whether churn behavior differs according to:

- Gender
- Customer tenure
- Monthly charges
- Contract type

Additionally, the project calculates the overall churn percentage to provide a high-level measure of customer attrition within the dataset.

The objective is not to predict future churn but to provide descriptive insights that can guide customer retention initiatives and highlight factors requiring further investigation.

BUSINESS BACKGROUND

The telecommunications industry operates in a highly competitive environment where customers have multiple service providers offering similar products, pricing plans, and service packages. As competition increases, customer retention becomes just as important as acquiring new customers. Even a small increase in customer churn can significantly impact recurring revenue, customer lifetime value, operational costs, and long-term business profitability.

Customer churn refers to the percentage of customers who discontinue a company's services within a given period. Every customer who leaves represents not only a loss of recurring revenue but also the additional expense of acquiring a replacement customer through marketing campaigns, promotional offers, and onboarding activities. Consequently, understanding why customers leave has become a key business priority for telecom organizations.

The dataset used in this project contains customer demographic information, account details, subscribed telecom services, contract information, billing characteristics, payment methods, tenure, and customer churn status. These variables provide an opportunity to investigate whether certain customer segments exhibit higher tendencies to discontinue their services.

Instead of focusing on predictive modeling, this project adopts an Exploratory Data Analysis (EDA) approach to uncover patterns and relationships within historical customer data. The objective is to identify observable trends that may help business stakeholders understand customer behavior and support future customer retention strategies.

The insights generated from this analysis can assist business teams in identifying high-risk customer groups, evaluating existing contract structures, assessing pricing strategies, and developing targeted retention initiatives. The project therefore serves as an analytical foundation for more advanced customer churn prediction and customer segmentation initiatives in future phases.

PROBLEM STATEMENT

Customer churn directly affects the financial performance and operational stability of subscription-based businesses. High customer attrition results in reduced recurring revenue, increased customer acquisition costs, and lower overall customer lifetime value. Understanding the factors associated with customer churn is therefore essential for designing effective retention strategies.

The primary objective of this project is to explore customer churn behavior using historical telecom customer data and identify characteristics that distinguish customers who discontinued their services from those who remained with the company.

The project investigates customer churn by examining multiple business dimensions available within the dataset, including customer demographics, account tenure, billing information, and contract type. Through exploratory analysis, the project seeks to identify meaningful relationships between these variables and customer churn without making predictive assumptions.

Rather than answering a single business question, the analysis aims to provide a broader understanding of customer behavior that can support future decision-making by business managers, customer success teams, and marketing departments. The findings from this exploratory analysis can also serve as a foundation for developing predictive churn models in subsequent analytical projects.

DATASET DOCUMENTATION

Dataset Overview

The project utilizes the **WA_Fn-UseC_-Telco-Customer-Churn.csv** dataset, which contains customer-level information for a telecommunications company. Each row represents a unique customer account, while each column captures a specific customer attribute related to demographics, subscribed services, account information, billing details, or churn status.

The dataset contains **7,043 customer records** distributed across **21 variables**.

The target variable used throughout the analysis is **Churn**, which identifies whether a customer has discontinued the telecom service.

Dataset Structure

The uploaded dataset contains the following verified columns:

- customerID
- gender
- SeniorCitizen
- Partner
- Dependents

- tenure
- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies
- Contract
- PaperlessBilling
- PaymentMethod
- MonthlyCharges
- TotalCharges
- Churn

These variables collectively describe customer demographics, subscribed telecom services, billing information, payment preferences, contract characteristics, and customer retention status.

Target Variable

The primary variable analyzed throughout the project is:

Churn

Possible values observed in the dataset:

- Yes
- No

This variable represents whether the customer remained with the telecom company or discontinued the service.

Business Relevance of Key Variables

Several variables play an important role in understanding customer churn and were utilized during the exploratory analysis.

Customer Tenure

Represents the duration of a customer's relationship with the telecom company. Tenure is frequently used to assess customer loyalty and long-term retention.

Monthly Charges

Represents the recurring monthly amount charged to each customer. Comparing monthly charges across churned and retained customers helps identify potential pricing-related patterns.

Total Charges

Represents the cumulative amount paid by a customer throughout their relationship with the company. Since this field contained non-numeric values, it required preprocessing before analysis.

Contract

Identifies the type of customer contract. Contract duration is an important business variable because longer contractual commitments often influence customer retention.

Gender

Represents customer gender and was analyzed to determine whether churn behavior differs across demographic groups.

Dataset Granularity

Each record represents one individual telecom customer.

No aggregation or summarization was performed before the analysis. All visualizations and calculations are based on customer-level observations after the data cleaning process.

DATA MODEL

The uploaded project uses a **single flat CSV dataset** as the source of analysis.

No relational database schema, star schema, snowflake schema, or dimensional data model was implemented within the uploaded project.

Since the project is based entirely on one dataset, all analytical operations are performed directly on the imported DataFrame after data cleaning.

The workflow follows the sequence below:

CSV Dataset

↓

Pandas DataFrame

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Data Validation

↓

Data Cleaning

↓

Exploratory Data Analysis

↓

Business Insights

↓

Business Recommendations

This simplified analytical architecture is appropriate for an exploratory data analysis project where the objective is to investigate relationships within a single customer dataset rather than integrating multiple business data sources.

DATA QUALITY ASSESSMENT

Before performing any analysis, the dataset underwent an initial quality assessment to verify its suitability for exploratory analysis.

The assessment focused on four primary areas:

Dataset Structure Validation

The dataset was initially inspected using the head() and info() functions to verify the available columns, record count, and data types. This preliminary inspection confirmed the overall structure of the dataset and identified columns requiring additional preprocessing.

Missing Value Assessment

Missing values across all columns were evaluated using the isnull().sum() function.

The analysis revealed that the TotalCharges column required additional attention because some records contained values that could not be interpreted as valid numeric data.

Data Type Validation

The project verified the data type of the TotalCharges column before beginning the cleaning process. This validation step confirmed that the column required conversion to a numeric format before numerical analysis could be performed.

Target Variable Verification

The distribution of the Churn variable was examined using value_counts() to verify the available categories and understand the overall balance between churned and retained customers before generating visualizations.

The data quality assessment established a clear understanding of the dataset prior to transformation and ensured that subsequent analyses were performed using validated and appropriately formatted data.

DATA CLEANING STRATEGY

Data cleaning was performed to improve the quality and reliability of the dataset before conducting exploratory analysis. Each transformation implemented in the Python script addressed a specific data quality issue identified during the initial assessment.

The first cleaning step involved converting the TotalCharges column from a text data type to a numeric format using the pd.to_numeric() function with the errors='coerce' parameter. This approach automatically converted invalid or non-numeric values into missing values (NaN) rather than causing execution errors.

Following the conversion, the dataset was re-evaluated to identify records containing missing values in the TotalCharges column. Since this variable was used in numerical analysis, incomplete records were removed using the dropna() function to ensure that subsequent calculations and visualizations were based only on valid numeric data.

After removing invalid records, the cleaned dataset was verified by examining its updated dimensions and data types. This final validation confirmed that the dataset was suitable for exploratory analysis and that the numerical variables required for visualization and business interpretation were correctly formatted.

The cleaning strategy adopted in this project prioritizes data integrity over record retention. Removing incomplete observations minimizes the risk of inaccurate calculations and ensures that business insights are derived from reliable customer information.

FEATURE ENGINEERING

Feature engineering involves creating new variables or transforming existing attributes to improve analytical depth and support more meaningful business insights. During the review of the uploaded project, no additional engineered features or calculated columns were implemented beyond the data cleaning process.

The analysis was performed using the original variables available in the dataset after converting the TotalCharges column into a numeric data type and removing records containing invalid values. Since the primary objective of the project was exploratory data analysis rather than predictive modeling, the existing variables provided sufficient information to investigate customer churn patterns.

No derived metrics such as customer lifetime value, average revenue per tenure, customer segments, risk scores, or tenure categories were created within the uploaded Python script. The project intentionally preserved the original dataset structure to ensure that the exploratory findings directly reflected the available customer information.

Although feature engineering was not required for this project, the cleaned dataset establishes a reliable foundation for future analytical enhancements such as customer segmentation, churn prediction models, and behavioral scoring systems.

TECHNICAL IMPLEMENTATION

The project was implemented entirely in Python using a structured exploratory data analysis workflow. The implementation follows a logical sequence beginning with data loading, dataset validation, data preparation, exploratory visualization, and business interpretation.

The analysis starts by importing the required Python libraries. Pandas is used for data manipulation and validation, while Matplotlib is used to create business-focused visualizations. These libraries provide all the functionality necessary for performing descriptive analysis on the customer dataset.

The telecom customer dataset is imported from the CSV file using the pd.read_csv() function. Once the dataset is loaded, the first few records are displayed to verify successful import and confirm the overall dataset structure.

Following data import, the project examines the dataset using the info() function to inspect column names, record counts, and data types. This preliminary assessment helps identify variables requiring additional preprocessing before analysis.

The project then evaluates missing values across all columns using isnull().sum(). During this validation process, the TotalCharges column is identified as requiring transformation because its data type prevents direct numerical analysis.

To resolve this issue, the TotalCharges column is converted into a numeric format using the pd.to_numeric() function with the errors='coerce' parameter. Invalid values are automatically converted into missing values (NaN), allowing the dataset to be cleaned without interrupting execution.

After conversion, rows containing missing values in TotalCharges are removed using the dropna() function. The cleaned dataset is subsequently validated by checking the updated dataset dimensions and confirming the revised data types.

Once the data preparation phase is complete, the project generates multiple visualizations using Matplotlib. These charts examine customer churn across different business dimensions and provide a visual representation of customer behavior.

Finally, the overall churn rate is calculated using the normalized frequency distribution of the Churn variable. This calculation provides a quantitative summary of customer attrition within the cleaned dataset and complements the graphical analysis.

The implementation emphasizes readability, simplicity, and reproducibility, making it suitable for both portfolio demonstration and future analytical enhancements.

EXECUTION WORKFLOW

The analytical workflow implemented in this project follows a structured sequence designed to ensure that the dataset is properly validated before business insights are generated. Each stage builds upon the previous step, creating a clear and reproducible analysis pipeline.

Step 1 – Library Import

The project begins by importing the required Python libraries for data manipulation and visualization.

Step 2 – Dataset Loading

The telecom customer dataset is imported from the CSV file into a Pandas DataFrame.

Step 3 – Initial Data Inspection

The first few records of the dataset are displayed to verify successful loading and understand the available variables.

Step 4 – Dataset Structure Validation

Column names, record counts, and data types are reviewed using the dataset information function.

Step 5 – Missing Value Assessment

The project evaluates missing values across all variables to identify potential data quality issues.

Step 6 – Data Type Conversion

The TotalCharges column is converted from a text field into a numeric variable to enable quantitative analysis.

Step 7 – Data Cleaning

Rows containing invalid or missing values in TotalCharges are removed to ensure reliable analysis.

Step 8 – Dataset Verification

The cleaned dataset is revalidated by reviewing data types and dataset dimensions.

Step 9 – Exploratory Data Analysis

Business-focused visualizations are generated to investigate customer churn across multiple dimensions, including customer demographics, tenure, pricing, and contract type.

Step 10 – Churn Rate Calculation

The overall customer churn rate is calculated from the cleaned dataset to provide a high-level business performance indicator.

Step 11 – Business Interpretation

The visualizations and churn statistics are interpreted to identify customer behavior patterns that may support customer retention strategies.

This structured workflow ensures that data validation precedes analysis, reducing the likelihood of misleading results and improving the reliability of business insights generated from the project.

BUSINESS QUESTIONS

The exploratory analysis was designed to investigate customer churn from multiple business perspectives. Rather than focusing solely on descriptive statistics, the project seeks to answer practical business questions that can support customer retention initiatives.

Business Question 1

What is the overall distribution of customer churn within the telecom customer base?

Business Objective

Determine the proportion of customers who remained with the company versus those who discontinued their services.

Business Importance

Understanding the overall churn distribution provides management with an initial assessment of customer retention performance and establishes a baseline for further investigation.

Business Question 2

Does customer churn vary based on gender?

Business Objective

Compare churn behavior between male and female customers to determine whether demographic differences exist.

Business Importance

Identifying demographic patterns can help evaluate whether customer retention initiatives should be customized for specific customer groups or applied uniformly across the customer base.

Business Question 3

How does customer tenure relate to customer churn?

Business Objective

Investigate whether the duration of a customer's relationship with the company influences the likelihood of discontinuing services.

Business Importance

Understanding the relationship between tenure and churn enables organizations to identify periods during the customer lifecycle where retention efforts may have the greatest impact.

Business Question 4

Is there a relationship between monthly charges and customer churn?

Business Objective

Examine whether customers paying higher monthly charges exhibit different churn behavior compared to customers with lower monthly charges.

Business Importance

Pricing strategy is a critical business decision. Understanding the relationship between monthly charges and customer retention can support pricing optimization and customer value management.

Business Question 5

How does contract type influence customer churn?

Business Objective

Analyze customer churn across different contract types to determine whether contractual commitments are associated with customer retention.

Business Importance

Contract structure directly affects customer commitment and revenue stability. Insights from this analysis can help evaluate whether encouraging longer-term contracts may contribute to reduced customer attrition.

The answers to these business questions are presented in the subsequent Business Analysis section, where each visualization is interpreted from a business perspective rather than simply describing graphical patterns.

BUSINESS ANALYSIS

The business analysis phase focuses on interpreting the exploratory data analysis results from a business perspective rather than merely describing the visualizations. Each analysis aims to explain how the observed customer behavior can influence operational decisions, customer retention strategies, and long-term business performance.

Analysis 1: Customer Churn Distribution

Business Objective

To understand the overall distribution of customers who remained with the company compared to those who discontinued their services.

Analysis Performed

A bar chart was generated to visualize the frequency of customers belonging to each churn category. Additionally, the overall churn rate was calculated using the normalized distribution of the target variable.

Business Interpretation

The analysis indicates that the majority of customers remained with the company, while a smaller proportion discontinued their services. Although retained customers represent the larger customer segment, the presence of a significant churn population highlights an important business concern that requires continuous monitoring.

Customer churn directly impacts recurring revenue, customer lifetime value, and operational efficiency. Even when the overall retention rate appears favorable, losing existing customers increases acquisition costs because attracting new customers is generally more expensive than retaining existing ones.

The calculated churn rate provides management with a high-level performance indicator that can be monitored over time to evaluate the effectiveness of customer retention strategies.

Business Impact

A measurable churn rate allows business leaders to assess the health of the customer base and identify whether retention initiatives require further investment. Monitoring this KPI over time also supports performance benchmarking and strategic planning.

Business Recommendation

The organization should establish periodic monitoring of customer churn and investigate customer segments contributing disproportionately to overall customer attrition. Continuous monitoring enables early identification of unfavorable trends before they significantly affect business performance.

Analysis 2: Churn by Gender

Business Objective

To determine whether customer churn differs between male and female customers.

Analysis Performed

A grouped bar chart was created to compare churn status across gender categories.

Business Interpretation

The visualization suggests that both male and female customers experience customer churn, with no substantial imbalance observed between the two demographic groups. This indicates that gender alone does not appear to be a primary factor influencing customer retention within the analyzed dataset.

From a business perspective, this observation suggests that customer retention strategies should prioritize behavioral, contractual, and financial characteristics rather than relying solely on demographic segmentation.

Although demographic variables remain useful for customer profiling, the available evidence indicates that gender should not be considered a primary driver of churn within this exploratory analysis.

Business Impact

Avoiding unnecessary demographic segmentation allows organizations to focus resources on variables that are more closely associated with customer attrition, improving the efficiency of retention campaigns.

Business Recommendation

Future customer retention initiatives should emphasize customer behavior, service usage, pricing, and contractual characteristics instead of designing gender-specific retention strategies.

Analysis 3: Customer Tenure vs Customer Churn

Business Objective

To examine the relationship between the length of the customer relationship and customer churn.

Analysis Performed

A histogram was generated to compare customer tenure across churned and retained customers.

Business Interpretation

The tenure distribution demonstrates that customer retention varies across different stages of the customer lifecycle. Customers with shorter service durations appear more frequently among churned customers, while longer-tenure customers are generally better represented within the retained customer population.

This observation suggests that customer loyalty strengthens as customers continue using the service over time. Customers who remain with the organization for extended periods may have established greater dependence on the services offered, making them less likely to switch providers.

From a business perspective, the early months of the customer lifecycle represent a critical period during which customer engagement, onboarding experience, and service quality can significantly influence long-term retention.

Business Impact

Customer acquisition investments may not generate expected long-term returns if newly acquired customers discontinue services shortly after joining. Improving early customer experiences can therefore increase customer lifetime value while reducing acquisition costs.

Business Recommendation

Develop structured onboarding programs for new customers, monitor early customer satisfaction, proactively resolve service issues during the initial months, and implement targeted engagement campaigns aimed at improving customer retention during the early stages of the customer lifecycle.

Analysis 4: Monthly Charges vs Customer Churn

Business Objective

To evaluate whether monthly billing amounts exhibit any relationship with customer churn.

Analysis Performed

A box plot was generated to compare the distribution of monthly charges for churned and retained customers.

Business Interpretation

The visualization indicates variation in monthly charges between retained and churned customers. This suggests that pricing may influence customer retention behavior, although pricing alone should not be interpreted as the sole reason customers discontinue services.

Higher monthly charges may increase customer expectations regarding service quality, customer support, and perceived value. When customer expectations are not fully met, dissatisfaction may contribute to higher churn tendencies.

Conversely, customers receiving greater value from premium services may continue using the service despite higher monthly charges. Therefore, monthly pricing should always be evaluated alongside service quality and customer satisfaction.

Business Impact

Pricing strategies directly affect revenue generation and customer retention. Understanding customer sensitivity to monthly charges allows organizations to optimize pricing structures without negatively impacting customer loyalty.

Business Recommendation

Organizations should regularly evaluate pricing competitiveness, improve communication regarding service value, and consider targeted promotional offers for customers who may be at higher risk of discontinuing services due to billing concerns.

Analysis 5: Contract Type vs Customer Churn

Business Objective

To investigate the relationship between customer contract duration and customer churn.

Analysis Performed

A grouped bar chart was created to compare churn behavior across different contract types.

Business Interpretation

The visualization demonstrates that customer churn varies across contract categories. Customers enrolled in month-to-month contracts exhibit noticeably higher churn compared to customers with one-year or two-year contractual commitments.

Longer-term contracts naturally encourage customer retention by increasing customer commitment and reducing the frequency of provider switching. In contrast, month-to-month customers possess greater flexibility to discontinue services whenever they become dissatisfied or receive more attractive offers from competitors.

Among all variables explored within this project, contract type appears to demonstrate one of the strongest observable relationships with customer churn.

Business Impact

Contract duration directly influences revenue stability, customer lifetime value, forecasting accuracy, and long-term business planning. Increasing the proportion of customers enrolled in long-term contracts may significantly improve customer retention performance.

Business Recommendation

The organization should encourage month-to-month customers to transition toward longer-term contractual agreements by offering loyalty incentives, discounted pricing, bundled service packages, or additional value-added benefits that strengthen long-term customer commitment.

KEY INSIGHTS

The exploratory analysis identified several important observations regarding customer churn within the telecom customer dataset.

The majority of customers remain with the company; however, the calculated churn rate indicates that customer attrition continues to represent an important business concern requiring continuous monitoring.

Gender does not appear to be a significant differentiating factor in customer churn. Customer retention strategies are therefore likely to achieve greater effectiveness when based on behavioral and service-related characteristics rather than demographic segmentation.

Customer tenure demonstrates a meaningful relationship with customer retention. Customers with shorter relationships appear more susceptible to discontinuing services, highlighting the importance of improving customer engagement during the early stages of the customer lifecycle.

Monthly charges exhibit observable variation between churned and retained customers, suggesting that pricing and perceived service value may influence customer retention decisions.

Contract type shows one of the strongest observable relationships with customer churn. Customers enrolled in month-to-month contracts appear considerably more likely to discontinue services than customers with longer contractual commitments.

Overall, the analysis indicates that customer retention is influenced more by account characteristics and customer lifecycle factors than by basic demographic attributes.

BUSINESS RECOMMENDATIONS

Based on the exploratory analysis conducted in this project, several practical business recommendations can support customer retention and improve long-term customer value.

The organization should prioritize retention initiatives for customers enrolled in month-to-month contracts by offering attractive upgrade options to one-year or two-year agreements. Long-term contractual commitments can improve customer stability while increasing revenue predictability.

New customer onboarding should receive greater organizational attention. Since shorter-tenure customers exhibit higher churn tendencies, structured onboarding programs, proactive customer support, and regular engagement during the initial months may improve long-term retention.

Pricing strategies should be periodically reviewed to ensure customers perceive appropriate value relative to the services received. Promotional offers, personalized discounts, or bundled service packages may improve customer satisfaction among customers with relatively higher monthly charges.

Customer churn should be monitored as a recurring business KPI rather than a one-time analytical exercise. Regular reporting enables organizations to evaluate whether retention initiatives are producing measurable improvements over time.

Finally, this exploratory analysis should be considered the foundation for more advanced analytics initiatives. Future predictive models, customer segmentation techniques, and churn risk scoring systems can build upon the cleaned dataset and analytical understanding established through this project.

KPI DOCUMENTATION

The project calculates and analyzes the following verified business KPI:

Customer Churn Rate

Definition

The percentage of customers who discontinued telecom services within the analyzed dataset.

Business Purpose

Customer Churn Rate measures overall customer retention performance and provides a high-level indicator of customer attrition.

Calculation Implemented in the Project

Customer Churn Rate (%) =

(Number of Customers with Churn = "Yes" ÷ Total Number of Customers) × 100

The Python implementation calculates this KPI using the normalized frequency distribution of the Churn variable after completion of the data cleaning process.

Business Value

Customer Churn Rate serves as a strategic performance indicator that enables organizations to monitor customer retention trends, evaluate retention initiatives, estimate revenue stability, and identify periods requiring additional customer engagement efforts.

The next section will cover **Challenges Faced, Lessons Learned, STAR Interview Version, Resume Master Description, ATS Keywords, Skills Demonstrated, Future Enhancements, Portfolio Assets, and References**, completing the master documentation.

CHALLENGES FACED

Every data analytics project presents practical challenges that must be resolved before reliable insights can be generated. Throughout the development of this Telecom Customer Churn Investigation project, several technical and analytical challenges were encountered during data preparation and exploratory analysis. Addressing these issues was essential to ensure the quality and reliability of the final results.

Data Type Inconsistency

One of the primary challenges identified during the initial data inspection was the TotalCharges column. Although this attribute represents a numerical business metric, it was imported as a text (object) data type. This prevented direct numerical analysis and visualization.

To resolve this issue, the column was converted into a numeric data type using the pd.to_numeric() function with the errors='coerce' parameter. This approach safely converted invalid values into missing values rather than causing execution errors, allowing the data cleaning process to continue without interruption.

Handling Invalid Records

Following the conversion of the TotalCharges column, several records contained missing values due to invalid or blank entries. These incomplete observations could negatively affect statistical calculations and graphical analysis if retained.

The project addressed this challenge by removing records containing missing values in the TotalCharges column using the dropna() function. Although this slightly reduced the dataset size, it improved the overall reliability of the analysis by ensuring that only valid customer records were included.

Ensuring Data Integrity Before Analysis

Generating business insights from unvalidated data can lead to misleading conclusions. Therefore, an important challenge was ensuring that the dataset was thoroughly validated before performing exploratory analysis.

This was addressed through a structured validation process involving dataset inspection, verification of column data types, assessment of missing values, and confirmation of the target variable distribution. These validation steps established confidence in the dataset prior to visualization and interpretation.

Selecting Business-Relevant Visualizations

Another challenge involved selecting visualizations that effectively communicated customer behavior while remaining easy for business stakeholders to interpret.

Instead of producing a large number of charts, the project focused on visualizations directly related to the business objective. The selected charts examined customer churn from multiple perspectives, including overall churn distribution, demographic characteristics, customer tenure, monthly billing, and contract type. This ensured that every visualization contributed meaningful business insights rather than serving only descriptive purposes.

Translating Data into Business Insights

Creating visualizations alone does not provide business value. A significant analytical challenge involved interpreting the observed patterns in a manner that supports practical decision-making.

The project therefore emphasized business interpretation alongside technical analysis. Each visualization was evaluated to understand its potential implications for customer retention, pricing strategies, customer engagement, and contract management, ensuring that the analysis remained aligned with real-world business objectives.

LESSONS LEARNED

This project provided valuable experience in applying exploratory data analysis techniques to a real-world business dataset while reinforcing several important analytical principles.

The first lesson learned is that data quality assessment should always precede any form of analysis. Even when datasets appear complete, incorrect data types or invalid values can significantly influence analytical outcomes. Performing systematic validation before visualization improves both the accuracy and credibility of the results.

The project also demonstrated the importance of understanding the business context before interpreting analytical findings. Visual patterns become meaningful only when they are connected to operational challenges such as customer retention, pricing strategies, and service management. Analytical work should therefore focus on explaining business implications rather than simply presenting descriptive statistics.

Another important lesson was the value of maintaining a structured analytical workflow. Dividing the project into clearly defined stages—including data loading, validation, cleaning, visualization, and interpretation—improves reproducibility, simplifies debugging, and makes the project easier to understand for future users.

The project further reinforced the importance of communicating findings in business language. Business stakeholders are generally more interested in actionable insights and recommendations than technical implementation details. Translating analytical results into practical business recommendations significantly increases the value of a data analytics project.

Finally, this project highlighted that exploratory data analysis serves as an essential foundation for advanced analytics. Understanding customer behavior through descriptive analysis creates a stronger basis for developing predictive models, customer segmentation techniques, and business intelligence dashboards in future projects.

STAR INTERVIEW VERSION

Situation

A telecommunications company collects extensive customer information, including demographics, subscribed services, billing details, contract types, and customer tenure. However, the organization requires a better understanding of customer churn behavior in order to support customer retention initiatives and reduce revenue loss associated with customer attrition.

Task

My responsibility was to perform an exploratory data analysis using Python to investigate customer churn patterns, identify variables associated with customer attrition, prepare the dataset for analysis, and communicate business-oriented insights that could support customer retention strategies.

Action

I began by importing the telecom customer dataset into Python and conducting an initial assessment of its structure, data types, and missing values. During the validation process, I identified that the TotalCharges column required conversion from a text data type to a numeric format. I performed this conversion using pd.to_numeric() and removed invalid records containing missing values to ensure reliable analysis.

Following data preparation, I generated multiple business-focused visualizations examining customer churn across several dimensions, including overall churn distribution, gender, customer tenure, monthly charges, and contract type. I also calculated the overall customer churn rate and interpreted each analysis from a business perspective, emphasizing customer retention rather than simply describing graphical outputs.

Result

The project successfully identified several observable patterns associated with customer churn. The analysis suggested that contract type, customer tenure, and monthly charges exhibit meaningful relationships with customer attrition, while gender appears to have limited influence. The project demonstrated a complete exploratory data analysis workflow, producing actionable business recommendations that can support future customer retention initiatives and provide a strong analytical foundation for predictive churn modeling.

RESUME MASTER DESCRIPTION

Developed a Telecom Customer Churn Investigation project using Python to perform exploratory data analysis on over 7,000 customer records. Conducted comprehensive data validation and preprocessing by converting the TotalCharges column to a numeric format, handling invalid values, and ensuring data quality prior to analysis. Performed business-focused exploratory analysis using Pandas and Matplotlib to evaluate customer churn across demographic characteristics, tenure, monthly charges, and contract types. Calculated customer churn rate, interpreted analytical findings from a business perspective, and formulated actionable recommendations to support customer retention strategies. Demonstrated proficiency in data cleaning, exploratory data analysis, business interpretation, visualization, and analytical communication.

ATS KEYWORDS

Domain Keywords

- Telecommunications
- Customer Churn
- Customer Retention
- Customer Analytics
- Business Analytics
- Subscription Analytics
- Customer Behavior Analysis

Technical Keywords

- Python
- Pandas
- Matplotlib
- CSV Data Processing
- Data Cleaning

- Data Validation
- Data Preprocessing
- Exploratory Data Analysis
- Data Visualization
- Descriptive Analytics

Business Intelligence Keywords

- Customer Insights
- Business Reporting
- KPI Analysis
- Churn Rate Analysis
- Business Recommendations
- Trend Analysis
- Customer Segmentation Foundation
- Analytical Problem Solving

SKILLS DEMONSTRATED

This project demonstrates a combination of technical, analytical, and business-oriented competencies relevant to entry-level Data Analyst roles.

Technical Skills

- Python Programming
- Pandas
- Matplotlib
- CSV Data Processing
- Data Cleaning
- Data Type Conversion
- Missing Value Handling
- Exploratory Data Analysis
- Data Visualization

Analytical Skills

- Data Validation
- Business Problem Analysis

- Trend Identification
- Customer Behavior Analysis
- KPI Calculation
- Business Interpretation
- Analytical Reasoning
- Insight Generation

Business Skills

- Customer Retention Analysis
- Business Recommendation Development
- Business Communication
- Report Documentation
- Data Storytelling
- Decision Support

FUTURE ENHANCEMENTS

The current project focuses on exploratory data analysis and establishes a strong analytical foundation for future improvements. Several enhancements can extend the scope and business value of the project.

A predictive machine learning model can be developed using the cleaned dataset to estimate the probability of customer churn for individual customers. Such a model could enable proactive customer retention initiatives by identifying high-risk customers before they discontinue services.

Customer segmentation techniques can be incorporated to group customers based on behavioral characteristics, service usage, or billing patterns. These segments could support more personalized marketing campaigns and targeted retention strategies.

Interactive dashboards developed using Power BI or Tableau could provide business users with dynamic access to customer churn metrics, enabling continuous monitoring of retention performance and facilitating data-driven decision-making.

Additional statistical analyses, feature engineering techniques, and model evaluation metrics may also be integrated to improve analytical depth and provide more comprehensive business insights.

PORTFOLIO ASSETS

The uploaded project contains the following portfolio assets:

Source Code

- telecom_churn.py

Dataset

- WA_Fn-UseC_-Telco-Customer-Churn.csv

Project Report

- Telecom_Churn_Analysis_Report.pdf

Generated Visualizations

- Customer Churn Distribution
- Churn by Gender
- Customer Tenure vs Churn
- Monthly Charges vs Churn
- Contract Type vs Churn

These assets collectively demonstrate the complete exploratory data analysis workflow, from data preparation to business interpretation.

REFERENCES

Dataset

WA_Fn-UseC_-Telco-Customer-Churn.csv (included in the uploaded project)

Programming Language

Python

Verified Python Libraries

- Pandas
- Matplotlib

No external APIs, SQL databases, Power BI dashboards, Excel workbooks, machine learning frameworks, or additional third-party data sources were identified in the uploaded project.

Overall Assessment

From a hiring manager's perspective, this project demonstrates the core competencies expected from an entry-level Data Analyst:

- A structured approach to data validation and cleaning.
- Competence in Python-based exploratory data analysis.
- The ability to translate technical findings into business insights.
- Clear communication of recommendations supported by analytical evidence.

- An understanding of customer-centric business problems within the telecommunications domain.

It serves as a solid portfolio project showcasing practical analytical skills while providing a foundation for future enhancements such as predictive modeling, customer segmentation, and business intelligence reporting.